9. A network of n locations is represented as a graph. Write a C program that accepts the graph using an Adjacency Matrix, accepts a starting vertex, performs a graph traversal, displays the visit order, and ensures that a vertex is not processed repeatedly. Test it with connected and partially connected graphs.

Source code :-

```
#include <stdio.h>
int main()
{
    int a[20][20], visited[20] = {0}, queue[20];
    int n, start, front = 0, rear = 0;
    int i, j, v;
    printf("Enter number of vertices: ");
    scanf("%d", &n);
    printf("Enter adjacency matrix:\n");
    for (i = 0; i < n; i++)
        for (j = 0; j < n; j++)
            scanf("%d", &a[i][j]);
    printf("Enter starting vertex (0 to %d): ", n - 1);
    scanf("%d", &start);
    if (start < 0 || start >= n)
    {
        printf("Invalid starting vertex\n");
        return 0;
    }
    visited[start] = 1;
    queue[rear++] = start;
    printf("BFS visit order: ");
    while (front < rear)
    {
        v = queue[front++];
        printf("%d ", v);
        for (j = 0; j < n; j++)
        {
            if (a[v][j] == 1 && visited[j] == 0)
            {
                visited[j] = 1;
                queue[rear++] = j;
            }
        }
    }
    printf("\n");
    for (i = 0; i < n; i++)
    {
        if (visited[i] == 0)
            printf("Vertex %d is not reachable from vertex %d\n", i, start);
    }
    return 0;
}
```

Output :-

Test 1: connected graph (5 vertices)

```
Enter number of vertices: 5
Enter adjacency matrix:
0 1 1 0 0
1 0 0 1 0
1 0 0 1 1
0 1 1 0 0
0 0 1 0 0
Enter starting vertex (0 to 4): 0
BFS visit order: 0 1 2 3 4
```

Test 2: partially connected graph (6 vertices)

```
Enter number of vertices: 6
Enter adjacency matrix:
0 1 0 0 0 0
1 0 1 0 0 0
0 1 0 0 0 0
0 0 0 0 1 0
0 0 0 1 0 0
0 0 0 0 0 0
Enter starting vertex (0 to 5): 0
BFS visit order: 0 1 2
Vertex 3 is not reachable from vertex 0
Vertex 4 is not reachable from vertex 0
Vertex 5 is not reachable from vertex 0
```